Mandatory Assignment 1

GFS189, RTL959

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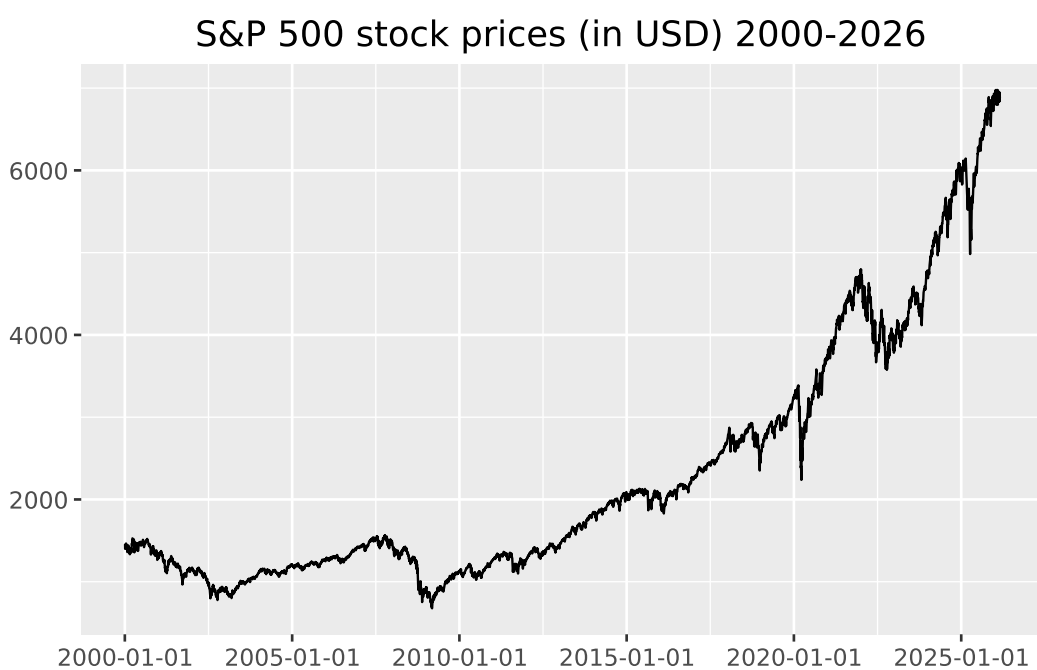
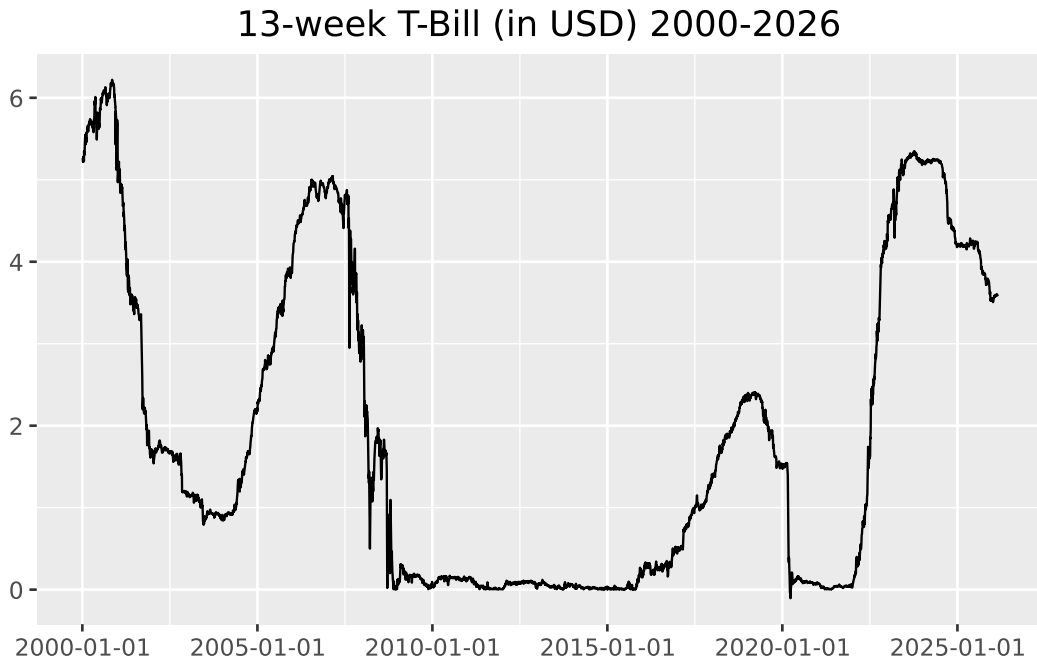
In this assignment we are investigating how different portfolio weights can lead to statistical errors based on the paper of Jobson and Korkie (1980) and Brandt (2010). The method to test this is to check how our portfolio weights deviate from the optimal in a single factor model market model which treats the calibrated parameters as the true values. we use this equation for portfolio weights:

$$\hat{\omega}(\bar{\mu}) = \arg\min_{\omega} \omega' \hat{\Sigma} \omega \quad \text{s.t.} \quad \omega' \iota = 1 \quad \text{and} \quad \omega' \hat{\mu} \geq \bar{\mu}$$

1

Firstly we gather data from S&P 500 index and the 13-week Treasury Bill rate from Yahoofinance using the Tidyfinance package. We se timeseries below in the figure. These figures primarily serve as a sanity check for our data and helps get better idea of their trends over time.





We now use the data to calculate these: Let $r_{m,t}$ be the monthly return and $r_{f,t}$ the risk-free rate.

1. Monthly Excess Return

$$\tilde{r}_{m,t} = r_{m,t} - r_{f,t}$$

2. Monthly Mean Excess Return

$$\hat{\mu}_m = \frac{1}{T}\sum_{t=1}^{T} \tilde{r}_{m,t}$$

3. Monthly Variance

$$\hat{\sigma}_m^2 = \frac{1}{T-1}\sum_{t=1}^{T} \left(\tilde{r}_{m,t} - \hat{\mu}_m\right)^2$$

4. Annualized Mean Excess Return

$$\hat{\mu}_{m,\text{ann}} = 12 \times \hat{\mu}_m$$

5. Annualized Variance

$$\hat{\sigma}_{m,\text{ann}}^2 = 12 \times \hat{\sigma}_m^2$$

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Monthly mean excess return:       0.006
Monthly variance:                 0.002
Annualized mean excess return:    0.071
Annualized variance:              0.023
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We see that the monthly mean excess return positive.

2

Now we construct a universe of N=50 synthetic assets which follow the single factor model: **Single Asset Return Process**

$$\tilde{r}_{i,t} = \beta_i \tilde{r}_{m,t} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2_{\varepsilon,i}), \quad i = 1, \dots, N$$

where $\tilde{r}_{i,t}$ denotes the excess return on asset i.

Parameter Draws

$$\beta_i \overset{iid}{\sim} \text{Uniform}(0.5,\ 1.5), \qquad \sigma_{\varepsilon,i} \overset{iid}{\sim} \text{Uniform}\left(\frac{0.10}{\sqrt{12}},\ \frac{0.30}{\sqrt{12}}\right)$$

True Monthly Parameters

$$\mu_i = \beta_i \hat{\mu}_m, \qquad \Sigma = \sigma^2_m \beta\beta' + \text{diag}(\sigma^2_\varepsilon)$$

where $\beta = (\beta_1, \dots, \beta_N)'$.

We see that the tangency portfolio has positive sharpe ratio and monthly return. ## Now we implement a function to compute the mean-variance efficient frontier and the tangency portfolio for a given covariance matrix and mean vector.

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Tangency portfolio Sharpe: 0.1335
Tangency mu (monthly):     0.0063
Tangency sigma (monthly):  0.0473
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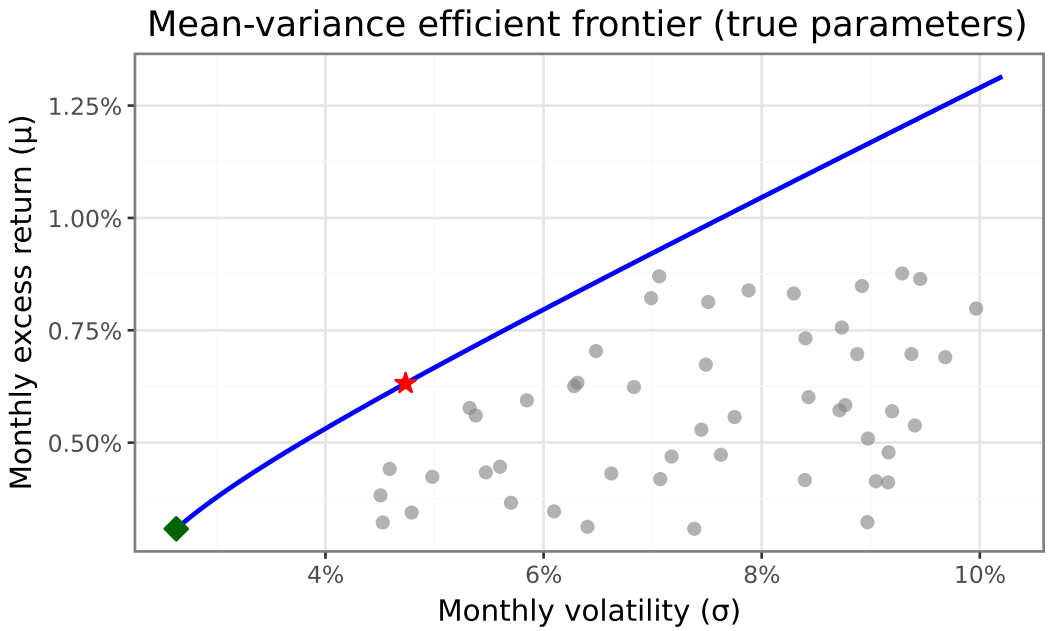
The figure shows that, as we would expect, that holding individual stocks is dominated by a combination of different stocks. This can be seen by all the grey dots representing the stocks lies below the efficient frontier.

3

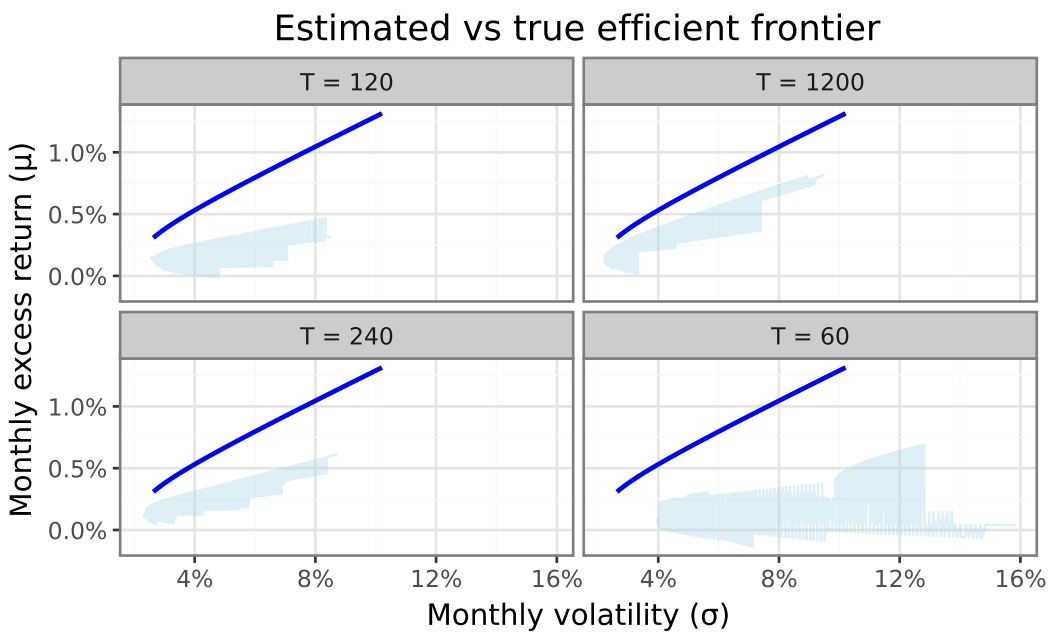
Now we implement a function to compute the mean-variance efficient frontier and the tangency portfolio for a given covariance matrix and mean vector.



ɟency portfolio. Green diamond: minimum variance portfolio. Grey dots: individual assets.

4

Now we show how estimation erros can affect the shape of the efficient frontier and what determines the severity of this.



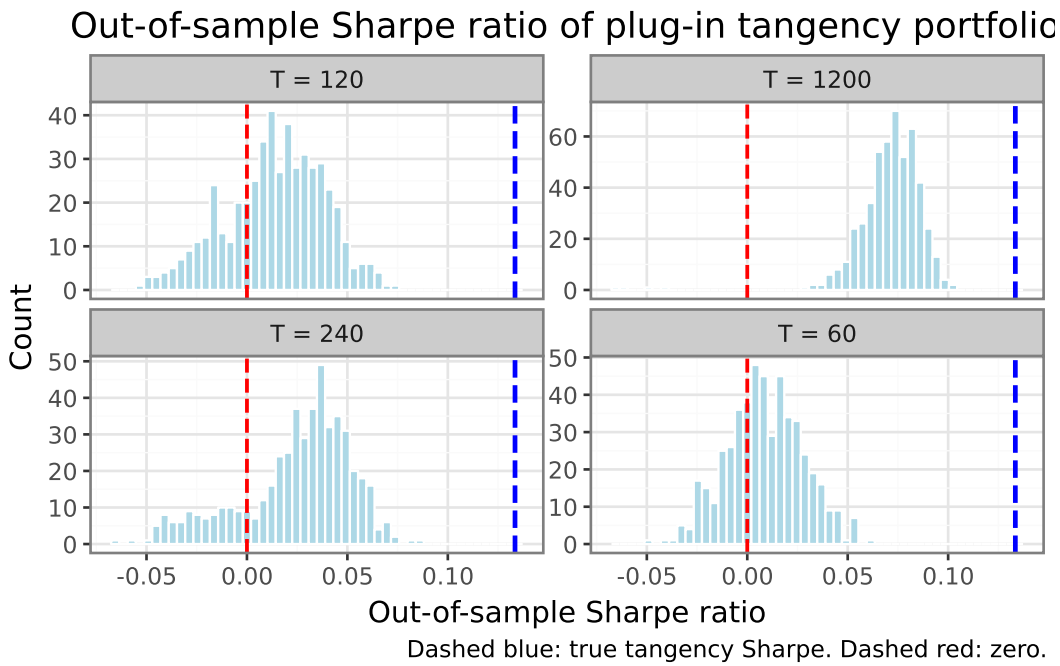
Light blue: plug-in frontiers. Dark blue: true frontier.

We see that as T increases(sample size) then the plug-in estimated frontiers goes towards the efficient frontier. The meaningful take away being that even with a big sample size of (T=1200) 100 years the plug in estimations suffers greatly from estimation error. The key drivers of estimation error are: 1. Sample size: greater sample size leads to converges to the efficient frontier 2. Number of assets:

when N is relatively big to T leads to poor estimation 3. Asset correlation: More correlation leads to poor estimation since the series becomes unstable Motivating the use of shrinkage estimators or factor models.

5

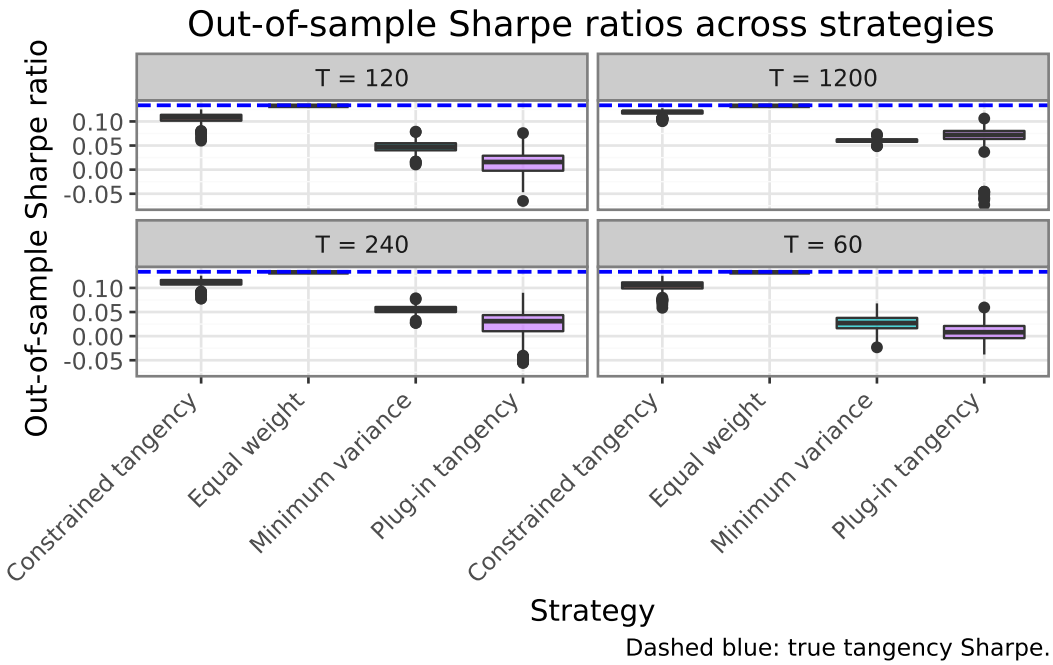
We now use simulations to see evaluate the out-of-sample performance of the plug-in tangency portfolio relative to the optimum. We use the sharpe ratio as the evaluations criteria since this is widely used and supported in the literature.



We see that for all simulations that the plug in tangency portfolio performs worse that the optimum (denoted by the dashed blue line). There is a trend for when T grows the Sharpe ratio tends towards optimum as we saw in the previous exercise as well.

6

We evaluate three alternative portfolio strategies and compare them against the equal-weight benchmark ($\omega_{ew} = \frac{1}{N}\iota$) to assess their ability to mitigate estimation error. All strategies are evaluated using the true parameters μ and Σ across 500 simulations for $T \in 60, 120, 240, 1200$. The three alternative strategies are: (i) the minimum variance portfolio, which only requires $\hat{\Sigma}$ and avoids estimating expected returns entirely; (ii) the short-sell constrained tangency portfolio, which limits extreme positions arising from estimation error by restricting all weights to be non-negative; and (iii) the unconstrained plug-in tangency portfolio as a baseline.



The figure shows that the equal-weight portfolio achieves the highest and most stable Sharpe ratio across all sample sizes, confirming that naive diversification is difficult to beat out-of-sample. The constrained tangency portfolio performs well at low T, as the no-short-selling constraint dampens the most extreme estimation-error-driven positions. The minimum variance and plug-in tangency portfolios perform similarly, with both being sensitive to sample size. Notably, none of the strategies reach the true tangency Sharpe ratio (dashed blue line), highlighting the persistent cost of estimation uncertainty.

7 Conclusion

We conlude that estimation error can have a significant impact on the performance of mean-variance optimal portfolios, especially when the sample size is small relative to the number of assets. The plug-in tangency portfolio, which relies heavily on estimated parameters, can perform poorly out-of-sample due to estimation error. Alternative strategies such as equal-weighting and constrained optimization can help mitigate this issue, but they also come with trade-offs. Overall, our analysis underscores the importance of considering estimation risk in portfolio construction and the potential benefits of simpler or more robust strategies in practice.

8 References

Brandt, M. W. (2010). Chapter 5 - portfolio choice problems. In AÏT-SAHALIA, Y. and HANSEN, L. P., editors, *Handbook of Financial Econometrics: Tools and Techniques*, volume 1 of *Handbooks in Finance*, pages 269–336. North-Holland, San Diego.

Jobson, J. D. and Korkie, B. (1980). Estimation for markowitz efficient portfolios. *Journal of the American Statistical Association*, 75(371):544–554.